

Challenge (Habanero)

- Q1.** What is the GCF of 12 and 18 using factors?
(A) 1
(B) 2
(C) 3
(D) 6
(E) 9
- Q2.** A gamer has 24 hours of gameplay and wants to divide it into segments with the same duration. If the segments must be a factor of 24, what are the possible segment lengths? What is the LCM of the possible segment lengths 4, 6, 8?
(A) 24
(B) 48
(C) 96
(D) 120
(E) 144
- Q3.** What is the GCF of 20 and 30 using prime factorization?
(A) 2
(B) 5
(C) 10
(D) $2 \cdot 5$
(E) $2 \cdot 2 \cdot 5$
- Q4.** A coding app has 15 and 25 minute levels. What is the LCM of 15 and 25?
(A) 15
(B) 25
(C) 75
(D) 100
(E) 150
- Q5.** What is the GCF of 9 and 12 using factors?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 6
- Q6.** A data plan has 300 MB and 600 MB limits. What is the LCM of 300 and 600?
(A) 300
(B) 600
(C) 900
(D) 1200
(E) 1800
- Q7.** A scoreboard has 8 and 12 point rewards. What is the GCF of 8 and 12 using prime factorization?
(A) 2
(B) $2 \cdot 2$
(C) $2 \cdot 2 \cdot 2$
(D) $2 \cdot 2 \cdot 2 \cdot 2$
(E) 4
- Q8.** A game developer wants to create levels with 18, 24, and 30 minute durations. What is the LCM of 18, 24, and 30?
(A) 72
(B) 120
(C) 144
(D) 180
(E) 360

Open Ended Questions (FRQ)

- Q9.** A gamer has 48 hours of gameplay and wants to divide it into segments of 8 hours on the first day and x hours on the subsequent days. If the total hours are divided evenly among 6 days, find x . Use GCF and LCM concepts to justify your answer. Show all work and explain your reasoning.
- Q10.** A data analyst has two datasets with 36 and 48 MB of data, respectively. If the datasets need to be merged into a new dataset with the LCM of 36 and 48, how much storage space is required in total? Explain your steps and provide calculations. Justify why the LCM is necessary for merging the datasets.

Chef's Tips

- Check for understanding of GCF and LCM concepts
- Emphasize the importance of prime factorization
- Provide feedback on problem-solving strategies